

STEM SMART Single Maths 34 - Integrating Rational Functions >

Integrals of Standard Functions 2

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P3

Complete the table of integrals for these standard functions.

In these functions a and k are constants.

Function, $y(x)$	Integral, $\int y(x)\mathrm{d}x$
$ax^n, n \neq -1$	
$a \sin kx$	
$a \cos kx$	
$a \sec^2 kx$	
$a\mathrm{e}^{kx}$	
$\frac{a}{x}$	

Items:

- $-\frac{a}{x^2} + c$

$\frac{a}{k} \tan kx + c$

$\frac{a}{n+1} x^{n+1} + c$

$\frac{a}{k} \cos kx + c$

$-\frac{a}{k} \cos kx + c$

$\frac{a}{k} \mathrm{e}^{kx} + c$

$-\frac{a}{k} \sin kx + c$
- $\frac{a}{k} \sin kx + c$

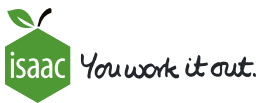
$a \ln |x| + c$

$a \ln x + c$

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Question deck:

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Integrating Rational Functions

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P3

Fill in the blanks to complete the notes on integrating rational functions.

In these functions a , p and q are constants.

A rational function is a function of the form $\frac{u(x)}{v(x)}$, where $u(x)$ and $v(x)$ are . To integrate a rational function, we start by splitting the function into and then use the integration rules below.

Function, $y(x)$	Integral, $\int y(x)dx$
$\frac{a}{x}$	
$\frac{a}{px + q}$	
$\frac{a}{(px + q)^2}$	
$a \frac{f'(x)}{f(x)}$	

Note that the last of these rules applies to function $f(x)$.

Items:

- any
- partial fractions
- polynomials
- $a \ln |x| + c$
- $\frac{a}{p} \ln |px + q| + c$
- $-\frac{a}{p(px + q)} + c$
- $a \ln |f(x)| + c$



Question

Partial Fractions 2ii

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Partial Fractions

Express $\frac{x-1}{x(x+1)}$ in partial fractions.

The following symbols may be useful: x

Part B

Integral

Hence find the exact value of $\int_1^2 \frac{x-1}{x(x+1)} dx$.

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Partial Fractions 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

Part A

Partial Fractions

Express $\frac{2+x^2}{(1+2x)(1-x)^2}$ in the form $\frac{A}{1+2x} + \frac{B}{1-x} + \frac{C}{(1-x)^2}$.

The following symbols may be useful: x

Part B

Integration

Hence find $\int_0^{\frac{1}{4}} \frac{2+x^2}{(1+2x)(1-x)^2} dx$ in exact form.

The following symbols may be useful: $\ln()$, $\log()$

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Partial Fractions 4ii

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Partial Fractions

Express $\frac{7-2x}{(x-2)^2}$ in the form $\frac{A}{x-2} + \frac{B}{(x-2)^2}$, where A , and B are constants.

The following symbols may be useful: x

Part B

Integral

Hence find the exact value of $\int_4^5 \frac{7-2x}{(x-2)^2} dx$.

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Integration by Substitution 4

Pre-Uni Maths for Sciences K3.4

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P3

Part A

Integrate $\frac{1}{b(x+a)}$

Find $\int_0^a \frac{1}{b(x+a)} dx$, where a and b are constants.

The following symbols may be useful: a , b , k , x

Part B

Integrate $\frac{x}{1+x^2}$

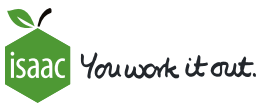
Find $\int_0^1 \frac{x}{1+x^2} dx$.

The following symbols may be useful: k , x

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Integration by Substitution 5

Pre-Uni Maths for Sciences K3.5

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P3

Part A

Integrate $\frac{x^3}{a^5+ax^4}$

Find $\int_a^{2a} \frac{x^3}{a^5+ax^4} dx$, where a is a constant.

The following symbols may be useful: a

Part B

Integrate $\tan \beta$

By writing $\tan \beta = \frac{\sin \beta}{\cos \beta}$, find $\int_0^{\frac{\pi}{4}} \tan \beta \, d\beta$.

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Algebraic Division 1ii

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Quotient and Remainder

Find the quotient when $3x^3 - x^2 + 10x - 3$ is divided by $x^2 + 3$.

The following symbols may be useful: x

Give the remainder.

The following symbols may be useful: x

Part B

Integral

Hence find the exact value of

$$\int_0^1 \frac{3x^3 - x^2 + 10x - 3}{x^2 + 3} dx.$$

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Integration with Partial Fractions 4

Pre-Uni Maths for Sciences K5.4

Subject & topics: Maths | Calculus | Integration Stage & difficulty: Further A P2

Part A

Find A , B and C

Write the function $\frac{2z^2 - z - 3}{(z + 2)(z^2 - 2z - 1)}$ in the form $\frac{A}{z + 2} + \frac{B + Cz}{z^2 - 2z - 1}$.

Drag and drop the correct values in the expression below.

Items:

-5

-4

-3

-2

-1

0

1

2

3

4

5

Part B

Integrate

Hence find $\int_1^2 \frac{2z^2 - z - 3}{(z + 2)(z^2 - 2z - 1)} \, dz$.